Information Notes 2

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Discrete case

Problem setting

We discuss the same problem for discrete cases, where the alphabet $\mathcal{X} = \{1, 2, \cdots, K\}$ and $x \in \mathcal{X}$, with probability $p(i) = p(x = i) = p_i$. and for convenience, let $p_1 > p_2 > \cdots > p_K$.
Our goal is to solve

$$\min_{q(\hat{x}|x)} I(X, \hat{X}) + \frac{1}{\lambda} E(D(X, \hat{X})),$$

where the distortion function is the *Hamming Distortion*

$$D(X, \hat{X}) = \mathbf{1}_{X \neq \hat{X}}.$$

Optimal Conditional PDF

According to BA algorithm, we have

$$q(\hat{x}|x) = \frac{r(\hat{x}) e^{-\frac{1}{\lambda}\mathbf{1}_{x \neq \hat{x}}}}{Z(x)},\ r(\hat{x}) = P(\hat{X} = \hat{x}),\ Z(x) = \sum_i r(i) e^{-\frac{1}{\lambda}\mathbf{1}_{x \neq \hat{x}}}$$

denote $\alpha = e^{-\frac{1}{\lambda}}$, then

$$\begin{aligned} Z(x) &= \sum_i r(i) e^{-\frac{1}{\lambda}\mathbf{1}_{x \neq \hat{x}}} \\ &= r(x) + \alpha(1 - r(x)) \\ &= \alpha + (1 - \alpha) r(x). \end{aligned}$$

So,

$$q(\hat{x}|x) = \begin{cases} \dfrac{r(x)}{\alpha + (1-\alpha)r(x)}, & \hat{x} = x, \\ \\ \dfrac{\alpha r(\hat{x})}{\alpha + (1-\alpha)r(x)}, & \hat{x} \neq x. \end{cases}$$

If the condition x $\notin supp(\hat{x}) \Rightarrow r(x) = 0 \Rightarrow q(\hat{x}|x) = r(\hat{x})$, which aligns with our assumption that the abandoned signal does not carry any information.

Threshold Truncation

For consistency, we should have

$$r(\hat{x}) = \sum_x p(x)q(\hat{x}|x), \text{ for } \forall \hat{x} \in supp(\hat{x}).$$

for any fixed $\hat{x} = t$,

$$\begin{aligned} r(t) &= p(t)\frac{r(t)}{Z(t)} + \sum_{x \neq t} p(x)\frac{\alpha r(t)}{Z(x)} \\ \Rightarrow 1 &= \frac{p(t)}{Z(t)} + \sum_{x \neq t} \alpha \frac{p(x)}{Z(x)} \\ \text{denote } S \triangleq \sum_x \frac{p(x)}{Z(x)} \Rightarrow 1 &= \frac{p(t)}{Z(t)} + \alpha(S - \frac{p(t)}{Z(t)}) \\ \Rightarrow \frac{p(t)}{Z(t)} &= \frac{1-\alpha S}{1-\alpha} \triangleq C, \quad \forall t \in supp(\hat{x}) \end{aligned}$$

So,

$$\begin{aligned} Z(x) &= \frac{p(x)}{C} \\ &= \alpha + (1-\alpha)r(x) \\ \Rightarrow r(x) &= \frac{\frac{p(x)}{c} - \alpha}{1-\alpha} \\ &= \frac{p(x) - \alpha C}{(1-\alpha)C}. \end{aligned}$$

Thus, $r(x) > 0 \Leftrightarrow p(x) > \alpha C$, which means $supp(r) = \{i \mid p_i > \alpha C\}$.
Suppose at current step,

$$supp(r) = \{1, 2, \cdots, m\},$$

Let $S_m = \sum_{i=1}^{m} p_i$, then

$$\begin{aligned}
1 &= \sum_{i=1}^{m} r(i) \\
&= \sum_{i=1}^{m} \frac{\frac{p_i}{C} - \alpha}{1-\alpha} \\
&= \frac{\frac{S_m}{C} - m\alpha}{1-\alpha} \\
\Rightarrow \quad C_{m,\alpha} &\triangleq C = \frac{S_m}{1+(1-m)\alpha} \\
\Rightarrow \quad r(i) &= \begin{cases} \dfrac{p_i - \alpha\, C_{m,\alpha}}{(1-\alpha)C_{m,\alpha}}, & i \le m, \\ 0, & i > m. \end{cases}
\end{aligned}$$

This is somewhat like "reverse water filling", set a threshold $\alpha\, C_m$ first, select $p_i > \alpha\, C_m$, subtract the threshold and normalized the pmf. Now we only need to determine the max of $supp(r)$, i.e. m.
Denote truncation threshold $\tau_{m,\alpha} \triangleq \alpha\, C_{m,\alpha}$,
for a given α, m must satisfies:

$$p_m > \tau_{m,\alpha}\ ,\ p_{m+1} \le \tau_{m,\alpha}$$

So we can search m by the following algorithm:

Algorithm 1 Determination of support Size m

Pre-computation: Calculate prefix sums $S_m = \sum_{i=1}^{m} p_i$ for all $m \in \{1, \ldots, K\}$.
for $m = K, K-1, \ldots, 1$ **do**
    Compute the threshold $\tau_{m,\alpha}$ using current S_m:

$$\tau_{m,\alpha} = \frac{\alpha S_m}{1 + (m-1)\alpha}$$

    Check validity condition:
    if $p_m > \tau_{m,\alpha}$ **and** $p_{m+1} \leq \tau_{m,\alpha}$ **then**
        return m
    end if
end for

When $\lambda \uparrow$, $\alpha = e^{-\frac{1}{\lambda}} \uparrow$, threshold $\tau_{m,\alpha} = \dfrac{S_m}{\frac{1}{\alpha} + (m-1)} \uparrow$, and some indices got out of the support. The critical condition is $r(m) = 0$, which means

$$\begin{aligned}
p_m = \tau_{m,\alpha} &= \frac{\alpha S_m}{1 + (m-1)\alpha} \\
\Rightarrow \alpha_m &= \frac{p_m}{S_m - (m-1)p_m} = e^{-\frac{1}{\lambda_m}} \\
\Rightarrow -\frac{1}{\lambda_m} &= ln(\frac{p_m}{S_m - (m-1)p_m}) \\
\Rightarrow \lambda_m &= \frac{1}{ln(\frac{S_m}{p_m} - (m-1))}
\end{aligned}$$

So m is abandoned from the support when λ gets larger from 0 to reach threshold λ_m.

Markov process

Another form of the conditional PDF

Using the fact that $Z_x = \frac{p_x}{C}$,

$$q(\hat{x}|x) = \begin{cases} \frac{r(x)}{Z_x} = \frac{r(x)C}{p_x} = \frac{p_x - \tau}{(1-\alpha)p_x} \ , & \hat{x} = x, \\ \frac{\alpha r(\hat{x})}{Z_x} = \frac{\alpha C r(\hat{x})}{p_x} = \frac{\tau}{p_x} r(\hat{x}) \ , & \hat{x} \neq x. \end{cases}$$

Let

$$t_x = \frac{\tau}{p_x} = \frac{\alpha}{Z_x},$$

for $\hat{x} = x$,

$$\begin{aligned} q(\hat{x}|x) &= \frac{r(x)}{Z_x} \\ &= \frac{r(x) + \alpha(1 - r(x)) - \alpha(1 - r(x))}{Z_x} \\ &= \frac{Z_x - \alpha(1 - r(x))}{Z_x} \\ &= \frac{p_x - \alpha C + \alpha C r(x)}{p_x} \\ &= \frac{p_x - \tau + \tau r(x)}{p_x} \\ &= (1 - \frac{\tau}{p_x}) + \frac{\tau}{p_x} r(x) \end{aligned}$$

and for $\hat{x} \neq x$,

$$q(\hat{x}|x) = \frac{\tau}{p_x} r(\hat{x}).$$

So, we can rewrite $q(\hat{x}|x)$ in another form:

$$q_\lambda(\hat{x}|x) = (1 - t_x)\delta_x + t_x r_\lambda(\hat{x}),$$

where δ_x represents the point mass at x.

Markov Transition Kernel

We give some notations first: for $\lambda_1 < \lambda_2$, denote

$$\alpha_i = e^{-\frac{1}{\lambda_i}} \ ; \ m_i = \max_j(supp(X_i)) \ ; \ r_i = r_{\lambda_i} \ ; \ \tau_i = \tau_{m_i,\alpha_i} \ , \ i \in \{1,2\}.$$

So, we naturally have $m_2 < m_1 \ ; \ \tau_2 > \tau_1$.
We seek a "transition kernel" $K_{\lambda_2|\lambda_1}(\hat{x_2}|\hat{x_1})$ that only relies on $\hat{X}_1$ such that

$$q_{\lambda_2}(\hat{x_2}|x) = \sum_{\hat{x_1}} K_{\lambda_2|\lambda_1}(\hat{x_2}|\hat{x_1})\, q_\lambda(\hat{x_1}|x).$$

For $x \leq m_2 \Rightarrow x \in supp(\hat{X}_1) \bigcap supp(\hat{X}_2)$, so

$$q_k(\cdot|x) = (1 - t_{k,x})\delta_x + t_{k,x} r_k \text{ , where } t_{k,x} = \frac{\tau_k}{p_x},\ k \in \{1,2\}.$$

and for $x > m_2,\ q_2(\cdot|x) = r_2$, so we hope K satisfies:

$$Kr_1 = r_2 \ ; \ K\delta_x \text{ is well selected for } \forall x \leq m_2.$$

for $u > m_2$,

$$\begin{aligned} q_2(v|u) &= K\, q_1(v|u) \text{ (here q is a vector)} \\ \Rightarrow r_2 &= (1 - t_{1,u})K\delta_u + t_{1,u}Kr_1 \\ &= (1 - t_{1,u})K\delta_u + t_{1,u}r_2 \\ \Rightarrow K\delta_u &= r_2 \\ \Rightarrow K(v|u) &= r_2(v). \end{aligned}$$

for $u \leq m_2$, we want K also has a linear combination form same as q, which is

$$K(\cdot|t) = (1 - a_t)\delta_t + a_t r_2(\cdot) \ , \text{ where } a_t \text{ TBD.}$$

take $x \leq m_2$,

$$
\begin{aligned}
q_2(\cdot|x) &= Kq_1(\cdot|x) \\
\Rightarrow (1-t_{2,x})\delta_x + t_{2,x}r_2 &= (1-t_{1,x})K\delta_x + t_{1,x}Kr_1 \\
&= (1-t_{1,x})\left[(1-a_x)\delta_x + a_x r_2\right] + t_{1,x}r_2 \\
&= (1-t_{1,x})(1-a_x)\delta_x + \left[1-(1-t_{1,x})(1-a_x)\right] r_2 \\
\Rightarrow (1-t_{1,x})(1-a_x) &= 1-t_{2,x} \\
\Rightarrow a_x &= \frac{t_{2,x}-t_{1,x}}{1-t_{2,x}} \\
&= \frac{\tau_2-\tau_1}{p_x-\tau_1}.
\end{aligned}
$$

Now, we have the close-form for the transition kernel matrix $K \in \mathbb{R}^{m_2 \times m_1}$:

$$
K(v|u) = \begin{cases} \dfrac{p_u-\tau_2}{p_u-\tau_1}\mathbf{1}_{v=u} + \dfrac{\tau_2-\tau_1}{p_u-\tau_1}r_2(v)\ , & u \leq m_2 \\ \\ r_2(v)\ , & m_2 < u \leq m_1. \end{cases} \quad , v \in \{1, 2, \cdots, m_2\}.
$$

which only relies on statistics of $\hat{X}_1$ and $\hat{X}_2$. And thus we have proven that $\{\hat{X_{\lambda_i}}\}$ is a Markov process, where the pmf eventually becomes $p(\hat{x}=1)=1$ when $\lambda \geq \lambda^* = \dfrac{1}{ln(\frac{p_1}{p_2})}$.
$K(u|v)$:

$$
\left(\begin{array}{cccc|ccc}
\alpha_1+\beta_1 r_2(1) & \beta_2 r_2(1) & \cdots & \beta_{m_2} r_2(1) & r_2(1) & \cdots & r_2(1) \\
\beta_1 r_2(2) & \ddots & & \vdots & \vdots & & \vdots \\
\vdots & & \ddots & \vdots & \vdots & & \vdots \\
\beta_1 r_2(m_2) & \cdots & \cdots & \alpha_{m_2}+\beta_{m_2} r_2(m_2) & r_2(m_2) & \cdots & r_2(m_2)
\end{array}\right)
$$

where

$$
\alpha_i = \frac{p_i-\tau_2}{p_i-\tau_1};\ \beta_i = 1-\alpha_i = \frac{\tau_2-\tau_1}{p_i-\tau_1}.
$$

and each column's sum is $\alpha_i + \beta_i = 1$, so $K(u|v)$ is a stochastic matrix.